

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			[Total] momentum before [a collision] = [total] momentum after [a collision] (1) provided no external forces act (1)	2			2		
	(b)			$35 \times 2 = 70$ (1) $0.45 \times [-]10 = [-]4.5$ (1) $70 - 4.5 = 65.5$ (1) kg m/s (1) Award 2 marks for an answer of 74.5	1 1	1 1		4	2	
	(c)			Substitution: 65.5 (ecf) = $35.45 \times v$ (1) $v = 1.85$ [m/s] or 1.8477 [m/s] (1) Accept 1.8 [m/s] ecf on 74.5 gives $v = 2.1$ [m/s] so award 2 marks	1	1		2	2	
	(d)			$\frac{1}{2} \times 35 \times 2^2 = 70$ [J] OR $\frac{1}{2} \times 0.45 \times 10^2 = 22.5$ [J] (1) KE before = 92.5 [J] ecf (1) KE after = $\frac{1}{2} \times 35.45 \times 1.85^2$ (ecf) (1) = 60.66 [J] (1) Conclusion must be given to award full marks i.e. so KE not the same so disagree ecf on 2.1 [m/s] gives 78.17 [J]			4	4	4	
				Question 8 total	5	3	4	12	8	0